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DNA Methylation Dynamics Along the Normal–Adjacent Axis in Breast Cancer

A Quantitative Statistical and Machine Learning Approach

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ABSTRACT DELLA TESI

Acknowledgements

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Glossary

GEO

Gene Expression Omnibus

IDAT

Intensity Data files

Chapter 1

Introduzione

INTRODUZIONE DELLA TESI

Chapter 2

BIOLOGIA E MEDICINA

CAPITOLO BIOLOGICO / MEDICO

Chapter 3

Data Preparation and Exploratory Analysis

3.1 Methodological Objectives

This chapter establishes the methodological foundation for all subsequent analyses by addressing the construction, organization, and preliminary characterization of the DNA methylation datasets used in this thesis. Following the biological background and problem formulation introduced in Chapters 1 and 2, the present chapter acts as a bridge between domain-specific knowledge and quantitative modeling, ensuring that downstream analyses are grounded on a technically sound and well-understood data representation.

Genome-wide DNA methylation data are characterized by extremely high dimensionality, heterogeneous signal-to-noise ratios, and the coexistence of subtle biological effects with multiple sources of technical variability. In this context, premature application of statistical models or feature-selection procedures may lead to biased conclusions driven by artefacts rather than genuine biological structure. A careful examination of data quality, distributional properties, and internal consistency is therefore a necessary prerequisite for any reliable inference [1].

The primary objective of this chapter is to construct a coherent and reproducible representation of the available datasets and to assess their structural and statistical properties prior to any form of preprocessing or modeling. Specifically, this chapter aims to: (i) describe the organization of methylation matrices and associated phenotype information; (ii) characterize the global and local variability of methylation levels within each dataset; (iii) evaluate the degree of separation between Normal, Adjacent, and Tumor tissues using exploratory tools; and (iv) compare datasets at an inter-cohort level to assess consistency, heterogeneity, and potential limitations in signal transferability.

The scope of this chapter is deliberately restricted to exploratory and diagnostic analyses. No feature selection, predictive modeling, or formal statistical inference is performed at this stage. All analyses are intended to be descriptive and comparative, serving to highlight structural properties of the data and to identify potential technical or biological issues that must be addressed before proceeding further.

Methodologically, the chapter is organized around a unified exploratory framework applied consistently across all datasets. Intra-dataset analyses are used to investigate internal structure, variability patterns, and group-level behavior, while inter-dataset comparisons focus on assessing cross-cohort consistency and platform-related differences. Throughout the chapter, visualizations and summary metrics are employed as diagnostic tools rather than as decision-making instruments.

The results of this exploratory phase reveal both shared characteristics and dataset-specific behaviors, as well as clear limitations of working directly with raw or minimally processed methylation data. These observations motivate the need for a structured and rigorous data pre-processing pipeline, which is developed and justified in the following chapter. The present chapter therefore provides the conceptual and technical basis for the preprocessing choices adopted in Chapter 4 and for all subsequent modeling steps.

3.2 Dataset Construction, Metadata Integration, and Data Organization

This section provides the methodological framework for the selection, construction of the dataset, and the organisation process adopted in this study. It contextualises the data sources, representation choices, metadata management, and computational design principles that underpin the detailed implementation described in the following sections.

3.2.1 Data Sources and Study Design

All datasets analyzed in this thesis were obtained from the NCBI Gene Expression Omnibus (GEO) [2], which was selected as a unified data source to minimize heterogeneity in data formats, metadata conventions, and access mechanisms. GEO provides curated series-level accessions, stable sample identifiers, and standardized links to processed methylation matrices and sample-level annotations, making it a natural reference repository for large-scale DNA methylation studies.

The study design adopts a multi-cohort comparative framework focused on genome-wide DNA methylation in breast tissue. Three independent GEO series were selected for analysis: **GSE69914** [3], based on the Illumina HumanMethylation450 platform, and two more recent EPIC-based cohorts, **GSE225845** [4] and

GSE287331 [5]. Together, these datasets span normal/healthy, adjacent-normal, and tumor tissue states across two generations of Illumina methylation arrays (450K and EPIC), enabling both within-dataset characterization and cross-dataset comparisons.

Dataset selection was driven by strict inclusion criteria aimed at ensuring technical compatibility, biological relevance, and analytical feasibility. In particular, only studies providing processed genome-wide methylation matrices in terms of β -values were considered, allowing a uniform data representation and avoiding the need for low-level signal reprocessing from raw `.idat` intensity files. Raw `.idat` files store probe-level fluorescence intensities measured in the methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) channels, from which β -values are derived as a normalized ratio [6]:

$$\beta := \frac{\max(y^{(M)}, 0)}{\max(y^{(M)}, 0) + \max(y^{(U)}, 0) + \alpha}, \quad \beta \in [0, 1]. \quad (3.1)$$

where α is a small offset (typically 100) used to stabilize the denominator. Additional requirements included the use of Illumina HumanMethylation450 (450K) or MethylationEPIC (EPIC) platforms, the availability of multiple tissue classes within each cohort, and sufficiently large sample sizes to support robust statistical analysis.

3.2.2 Methylation Data Representation and Storage

In several datasets, the original authors applied established normalization procedures, including methods specifically designed to correct probe-type design bias such as BMIQ [7]. Reprocessing raw IDAT files would therefore not add information relevant to the objectives of this thesis, while substantially increasing technical complexity and reducing reproducibility. For these reasons, all analyses were conducted on the processed β -value matrices as distributed via GEO.

To support scalable analysis and efficient input/output operations, all working methylation matrices were converted to a standardized `sample` $\times$ `CpG` layout and stored in columnar Parquet format using single-precision floating-point encoding.

This choice is motivated by both numerical and computational considerations. Since β -values are strictly bounded in the interval $[0, 1]$ [6], and biologically meaningful methylation differences typically occur at magnitudes on the order of 10^{-2} – 10^{-3} , single-precision floating point (`float32`, machine $\varepsilon \approx 10^{-7}$) provides more than sufficient numerical accuracy. At the same time, it substantially reduces memory usage and I/O overhead compared to double precision.

This representation is consistent with recent large-scale genomics frameworks that process molecular features, including DNA methylation data, entirely in `float32` precision for efficiency and scalability [8].

3.2.3 Phenotype Tables and Label Harmonization

For each dataset, a dedicated phenotype table was constructed by extracting and parsing GEO sample-level metadata. These tables encode tissue labels, sample identifiers, and relevant clinical or technical covariates when available. Given the heterogeneity of original annotations across studies, tissue classes were mapped to harmonized numeric encodings to ensure consistency across cohorts while preserving dataset-specific attributes in separate fields.

A strict one-to-one correspondence between methylation matrices and phenotype tables was enforced. Samples lacking either methylation measurements or corresponding metadata were excluded to prevent inconsistencies and downstream misalignment. This design guarantees that all analyses operate on synchronized molecular and phenotypic representations and enables reliable stratification by tissue class in both intra- and inter-dataset comparisons.

3.2.4 Reproducibility and Computational Workflow

The overall data organization and computational design follow a multi-dataset analytical paradigm commonly adopted in recent large-scale DNA methylation studies, in which cohort-specific analyses are complemented by cross-dataset comparisons to explicitly evaluate robustness and generalizability, while simultaneously revealing cohort-dependent effects and limitations in signal transferability that motivate subsequent methodological refinement [9].

All dataset construction and organization steps were implemented using deterministic, script-based pipelines with explicit schema definitions and typed data representations. Particular care was taken to avoid unnecessary in-memory operations on high-dimensional methylation matrices, favoring streaming ingestion, chunked processing, and out-of-core transformations when required by dataset size.

The resulting data organization yields a set of standardized, self-contained datasets, each consisting of a genome-wide methylation matrix and an aligned phenotype table stored in efficient, portable formats. This design ensures full reproducibility of the data preparation process and provides a robust foundation for the preprocessing, exploratory analysis, and modeling steps developed in subsequent chapters.

3.3 Intra-Dataset Exploration and Visualization

This section presents a structured intra-dataset exploratory analysis aimed at characterising the internal statistical and biological organisation of each methylation cohort prior to any preprocessing or modelling step. Each dataset is analysed independently in order to assess data integrity, quantify variability patterns, and

Table 3.1: Overview of the GSE69914 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	Normal (BRCA1)	Tumor (BRCA1)
485,512	407	50	42	305	3	7

evaluate whether Normal, Adjacent, and Tumor tissues exhibit distinguishable epigenetic behaviour.

In the context of breast cancer, epigenetic deregulation is known to manifest through both global and local phenomena, including genome-wide hypomethylation, focal hypermethylation of tumour-suppressor regions, and the presence of pre-neoplastic “field defects” in histologically normal tissues proximal to tumours [10, 11, 12]. These mechanisms motivate an exploratory investigation focused not only on overt tumour-associated changes, but also on more subtle alterations in Normal and Adjacent samples, which are central to the biological hypothesis of this thesis.

This section therefore applies a unified exploratory framework to each dataset (GSE69914 [3.3.1], GSE225845 [3.3.2], GSE287331 [3.3.3]), systematically examining data quality, global methylation distributions, sample-level summaries, CpG-wise variability and instability, correlation structure, dimensionality reduction, and genomic context coverage. The use of a consistent analytical protocol enables direct comparison of intra-dataset behaviour while preserving cohort-specific characteristics.

3.3.1 Dataset GSE69914

Dataset overview GSE69914 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium HumanMethylation450 BeadChip, providing genome-wide coverage of approximately 4.8×10^5 CpG loci. The dataset comprises samples spanning the full Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

An overview of the sample composition and CpG coverage is reported in Table 3.1, including the distribution across tissue classes and hereditary-risk subgroups.

A limited subset of samples is associated with germline *BRCA1* mutations. *BRCA1*, in particular, encodes a key DNA repair protein, and pathogenic variants are known to substantially increase lifetime breast and ovarian cancer risk [13]. Multiple studies have shown that BRCA-associated breast tissues exhibit distinct baseline epigenetic profiles, reflecting inherited genomic instability rather than sporadic tumorigenesis [14, 15].

Since the primary objective of this thesis is to characterise early, field defect DNA methylation alterations along the Normal–Adjacent axis in sporadic breast cancer, samples carrying *BRCA1* mutations were excluded from the core exploratory and

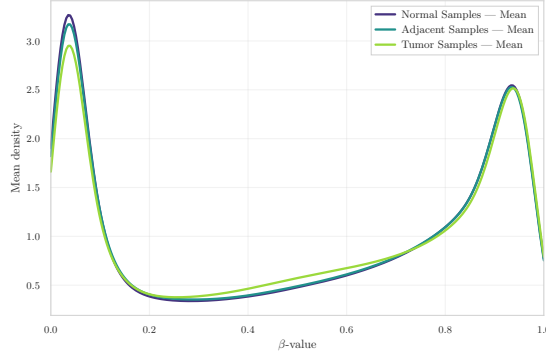


Figure 3.1: Group-wise mean β -value density in GSE69914.

preprocessing pipeline. Their inclusion would introduce a confounding hereditary epigenetic signal, potentially obscuring subtle methylation shifts in non-mutated normal and adjacent tissues.

Global methylation distributions To characterise the overall methylation landscape of the dataset, the distribution of β -values (fractional methylation in $[0,1]$) was first examined. The group-wise mean density curve (Figure 3.1) displays the characteristic *bimodal* profile of Illumina 450k arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. This pattern reflects the underlying biology of CpG regulation, where many loci tend to be either transcriptionally active (hypomethylated) or repressed (hypermethylated) [11, 16].

When comparing tissue groups, a clear gradient emerges. Tumor samples show a slightly flatter high- β peak and a broader low- β tail, consistent with the well-described phenomenon of global hypomethylation and increased heterogeneity in cancer [10]. Adjacent samples lie between Normal and Tumor, suggesting early epigenetic drift and subtle field defects occurring in histologically non-neoplastic tissue [12].

CpG-level instability and recurrent outliers To characterise locus-specific instability, the 200 CpG sites with the highest outlier burden across samples were examined. The heatmap of the top outlier loci (Figure 3.2a) indicates that epigenetic disruption is *not uniform*: specific CpG sites and specific samples display extreme deviations, rather than a diffuse genome-wide shift. Moreover, both directions of alteration are present — focal hypermethylation and focal hypomethylation. In cancer biology, hypermethylation can silence tumor-suppressor regions, whereas hypomethylation can derepress oncogenic pathways and weaken genomic stability, reflecting the classic “too much and too little methylation” behaviour of tumor genomes [10].

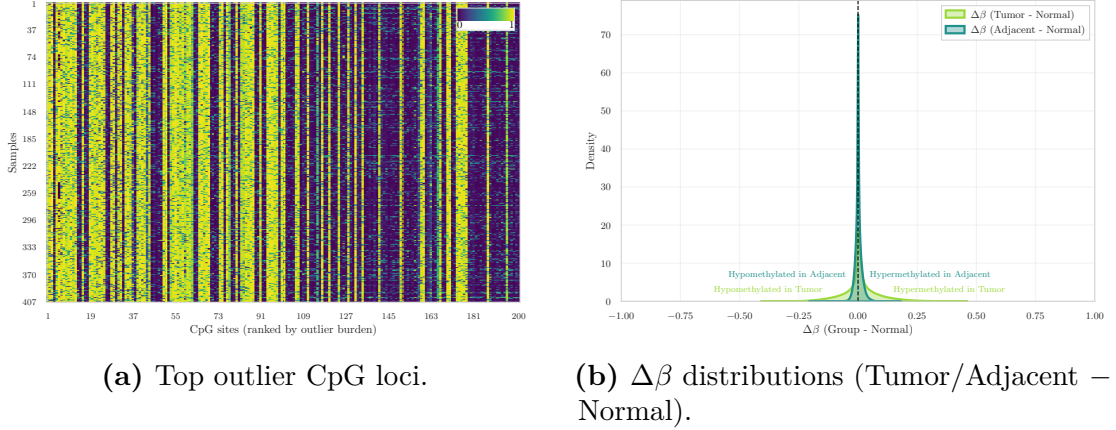


Figure 3.2: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE69914.

Complementary $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.2b) show a clear shift toward hypomethylation in Tumor samples and a subtler but detectable drift in Adjacent tissues. This provides evidence that epigenetic alterations emerge early in histologically normal tissue located near the tumor, supporting the field-cancerization model [12].

Variance and correlation structure To quantify within-group epigenetic variability, the distribution of CpG-wise variance across samples was examined. The density curves (Figure 3.3a) show that Tumor samples have markedly higher variance, reflecting increased epigenetic instability. In contrast, Normal and Adjacent samples display almost identical variance distributions that are tightly concentrated near zero. This indicates that, at the level of CpG-wise variability, Adjacent tissues do not yet exhibit detectable divergence from Normal samples.

A complementary perspective is provided by the sample correlation heatmap (Figure 3.3b), which shows the expected high level of pairwise similarity across all samples. This behaviour is typical of genome-wide DNA methylation data, where a large fraction of CpG sites is stable across individuals, resulting in consistently strong correlations [7]. Importantly, the heatmap also reveals that all samples are highly mutually correlated, with no sharp blocks or boundaries separating the three tissue labels. This confirms that global methylation structure is largely conserved across Normal, Adjacent and Tumor tissues, and that group differences are too subtle to be detected through sample-sample correlation alone.

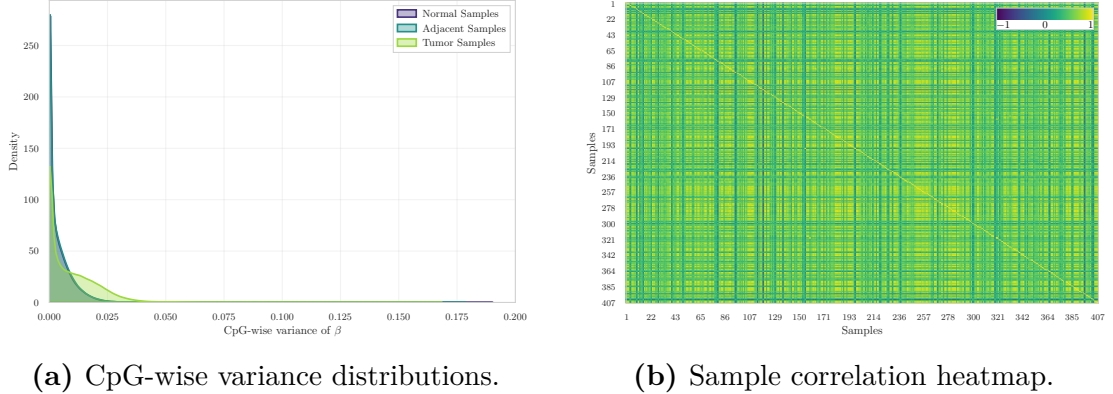


Figure 3.3: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE69914.

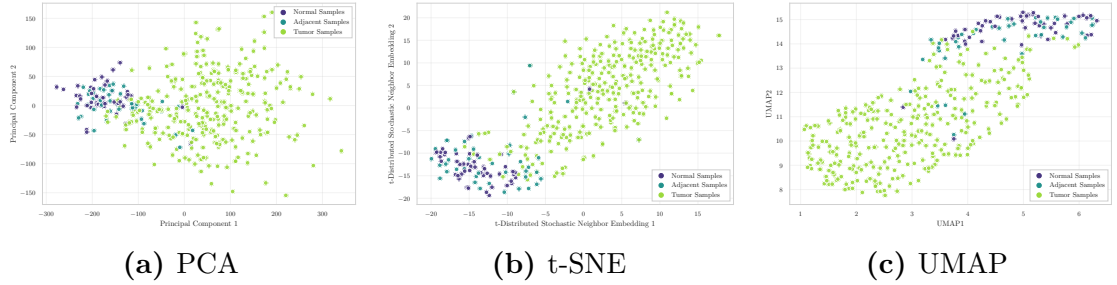


Figure 3.4: Low-dimensional embeddings samples in GSE69914.

Low-dimensional embeddings Dimensionality reduction was applied using Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) to visualise global similarities among samples (Figure 3.4). Across all three methods, Normal and Adjacent samples show substantial overlap, with no clear separation between these two groups. Tumor samples appear more dispersed, but only mildly shifted relative to the Normal/Adjacent cluster, and no sharp cluster boundaries emerge in any embedding.

These projections indicate that global methylation patterns alone do not provide strong discriminative structure among the three tissue states. This suggests that group differences are subtle at the whole-methylome level and are more effectively captured through locus-specific analyses rather than through global unsupervised embeddings.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* [17] manifest file, the standard Illumina resource

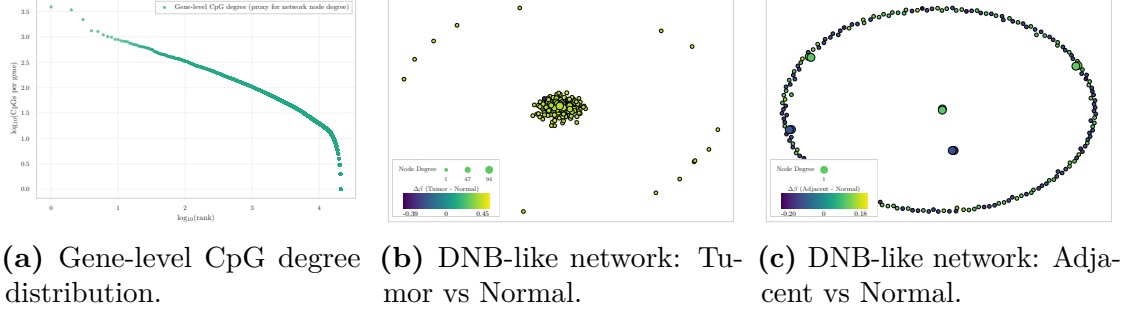


Figure 3.5: Dynamic-network proxy visualizations in GSE69914.

containing genomic coordinates, probe type, CpG island context (Island, Shore, Shelf, Open Sea), and gene-level annotations. Using this manifest, the dataset exhibits the expected non-uniform genomic distribution of CpG contexts: the largest fraction of probes maps to open-sea or non-island regions, followed by CpG islands, then north and south shores, and finally north and south shelves. This ordering mirrors the characteristic architecture of the HM450 arrays reported in the literature [18, 19]. The coverage observed is therefore consistent with the known genomic composition of the Illumina methylation arrays, supporting correct manifest alignment.

Dynamic-network proxies Although this analysis does not follow the full mathematical formalism of Dynamic Network Biomarkers (DNBs), the degree-based and network-level (Figure 3.5) provide a qualitative view of how methylation perturbations may organize at the network level. DNB theory, originally formulated for gene-expression and regulatory networks, predicts that near a critical transition a subset of components displays increased internal fluctuations and coordinated behaviour [20, 21, 22]. Here, this conceptual framework is adapted to CpG-level methylation patterns by examining the structure of CpG-gene degree distributions and the organization of $\Delta\beta$ -driven network layouts. This approach enables the exploration of whether tumour-associated instability manifests as locally coordinated methylation shifts at the network level, even without implementing the complete DNB pipeline.

The degree-rank plot (Figure 3.5a) displays a heavy-tailed distribution of CpG counts per gene, consistent with the heterogeneous hub-periphery structure typically observed in biological systems. In the Tumor vs Normal comparison (Figure 3.5b), a compact nucleus of nodes with moderately high degree is observed, accompanied by larger $\Delta\beta$ deviations. This pattern is consistent with a more perturbed and locally coordinated subset of CpG/gene units in tumor tissue, qualitatively compatible with increased network instability. In contrast, the Adjacent vs Normal network

Table 3.2: Overview of the GSE225845 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor
750,426	477	113	140	224

(Figure 3.5c) does not exhibit a comparable hub-like concentration and shows smaller, more diffuse deviations, suggesting that the adjacent tissue reflects only early or weakly organized perturbations.

Although these observations do not satisfy the full mathematical criteria for identifying a DNB module, they are conceptually consistent with the interpretation that tumor tissue occupies a more unstable network regime, whereas adjacent tissue remains closer to a stable (pre-transition) configuration.

Overall interpretation The exploratory analysis of GSE69914 reveals a consistent but overall subtle separation between the three tissue states. Normal and Adjacent samples exhibit highly similar global methylation profiles, variance distributions, correlation structure, and low-dimensional embeddings, indicating that large-scale methylome organisation remains largely conserved in Adjacent tissue. Across these global summaries, the two groups are so intermixed that no clear or marked distinction emerges between Normal and Adjacent samples when using whole-methylome descriptors. Tumor samples display the expected increase in heterogeneity and a shift toward hypomethylation; however, these differences remain modest at the whole-methylome level and do not produce clear separation in unsupervised embeddings.

Locus-specific analyses, however, highlight focused patterns of disruption. Outlier CpGs exhibit both hyper- and hypomethylation events, and $\Delta\beta$ distributions reveal a distinct hypomethylation bias in Tumor samples together with a milder but detectable shift in Adjacent tissue. Dynamic-network proxies further indicate a more compact and perturbed core in Tumor samples, whereas Adjacent tissue appears more diffuse and weakly structured.

Overall, GSE69914 constitutes a technically clean dataset with well-structured methylation patterns and biologically interpretable deviations. The results indicate that tumour-related alterations are primarily localized at specific CpG loci rather than reflected in global methylome reorganization. Within this framework, the Adjacent tissue exhibits subtle yet measurable deviations from the Normal baseline, consistent with previously described field-effect patterns in breast cancer epigenomics.

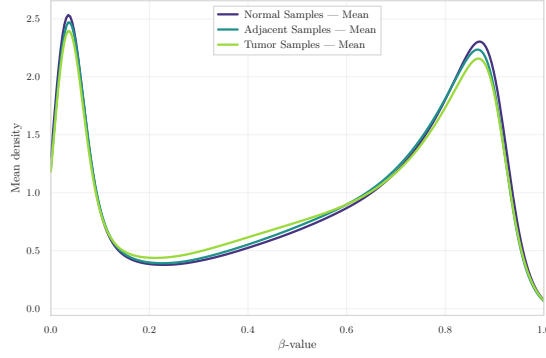


Figure 3.6: Group-wise mean β -value density in GSE225845.

3.3.2 Dataset GSE225845

Dataset overview GSE225845 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC BeadChip (850K), providing genome-wide coverage of more than 8.5×10^5 CpG loci. The dataset is part of the NCI-Maryland Breast Cancer Cohort and includes samples spanning the Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

An overview of the sample composition and CpG coverage after cohort harmonization is reported in Table 3.2, including the distribution across tissue classes.

The dataset further provides detailed demographic and clinical metadata (e.g., age at surgery, race, sex, and tissue descriptors), enabling controlled downstream analyses when required.

Global methylation distributions The group-wise mean β -value densities (Figure 3.6) display the expected bimodal configuration of EPIC arrays. As observed in GSE69914, the three tissue groups show a high degree of overlap across the entire β range.

Normal and Adjacent samples exhibit nearly indistinguishable density profiles. Tumor samples present only minimal deviations, characterised by a slight increase in density at low β values and a marginal attenuation of the high- β peak. At the whole-methylome level, no marked global redistribution of methylation levels is evident.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.7a) indicates that instability within this subset is predominantly directional. Recurrent outlier events are largely associated with very low β values, consistent with focal hypomethylation affecting a restricted set of CpG sites. The perturbation pattern therefore appears concentrated rather than diffuse, suggesting

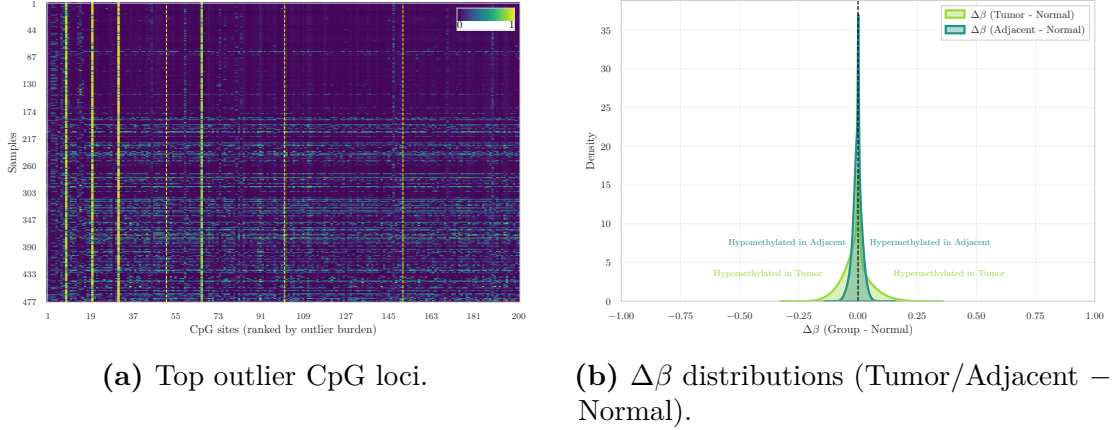


Figure 3.7: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE225845.

localized instability rather than a genome-wide redistribution.

The corresponding $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.7b) are sharply centered around zero, indicating that most loci remain stable. However, both comparisons exhibit a heavier negative tail relative to the positive side. Tumor samples display the most extended negative tail, consistent with stronger hypomethylation shifts, whereas Adjacent tissue shows a milder but still detectable asymmetry toward $\Delta\beta < 0$.

Variance and correlation structure The CpG-wise variance distributions (Figure 3.8a) indicate clear differences in variability across tissue groups. Normal samples exhibit highly concentrated variance values near zero, whereas Tumor samples display a substantially longer right tail, reflecting an increased proportion of CpG loci with elevated variability. Adjacent samples show a broader distribution than Normal tissues, with partial overlap with both Normal and Tumor profiles, but without forming a strictly intermediate pattern.

The sample correlation heatmap (Figure 3.8b) shows uniformly positive correlations across the dataset. Most pairwise values lie within a narrow high-correlation range, and no distinct block structures separating tissue classes are observed. This pattern indicates preservation of global methylation structure, with group differences emerging primarily from locus-specific variability rather than from large-scale shifts in overall methylome organization.

Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise global similarities among samples (Figure 3.9).

In PCA space, the three tissue groups largely overlap. Tumor samples show a

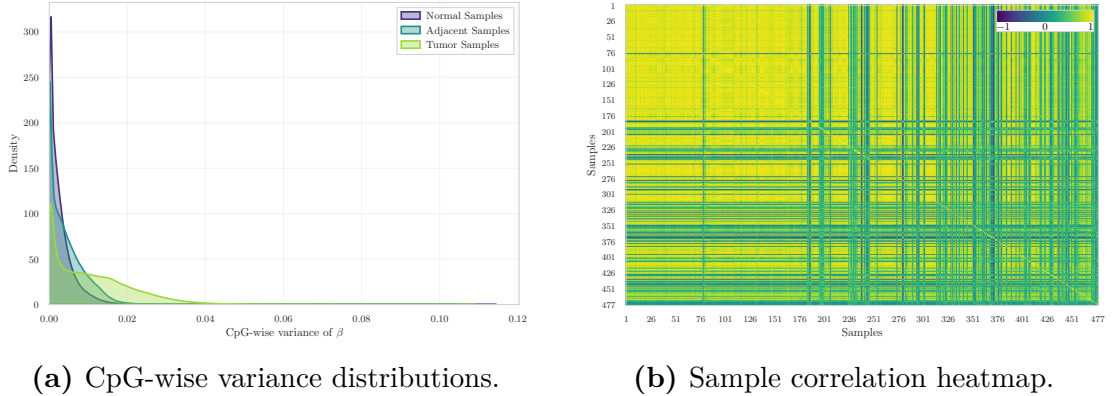


Figure 3.8: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE225845.

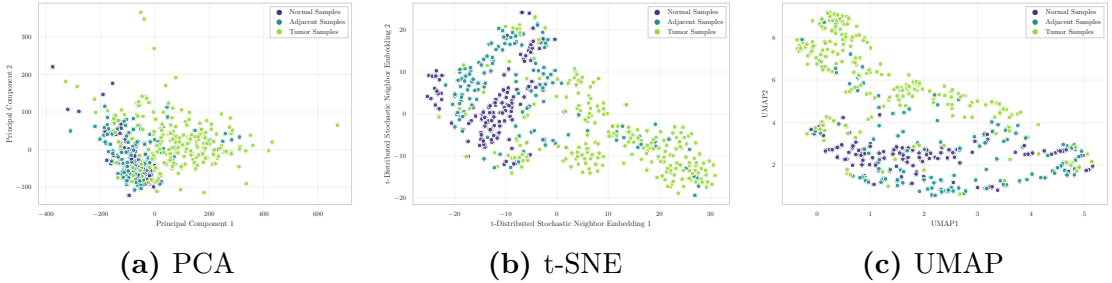


Figure 3.9: Low-dimensional embeddings samples in GSE225845.

slightly broader dispersion, particularly along the second principal component, but no clear linear separation is observed.

Nonlinear embeddings reveal additional structure. In t-SNE space, Tumor samples occupy a more extended region and appear more dispersed, whereas Normal and Adjacent samples remain partially intermixed. Several Tumor samples are positioned at the periphery of the embedding, consistent with increased heterogeneity.

The UMAP projection exhibits a similar configuration. Tumor samples span a wider area and extend toward higher values of the second UMAP axis, while Normal and Adjacent samples form relatively compact yet overlapping regions. Despite differences in spread, none of the methods yields sharply separated or isolated clusters.

Genome annotation coverage Consistent with the architecture of the EPIC platform, the genomic distribution of annotated CpG contexts is markedly unbalanced. The largest fraction of probes maps to open-sea or non-island regions, which constitute the dominant category. CpG islands represent the second most

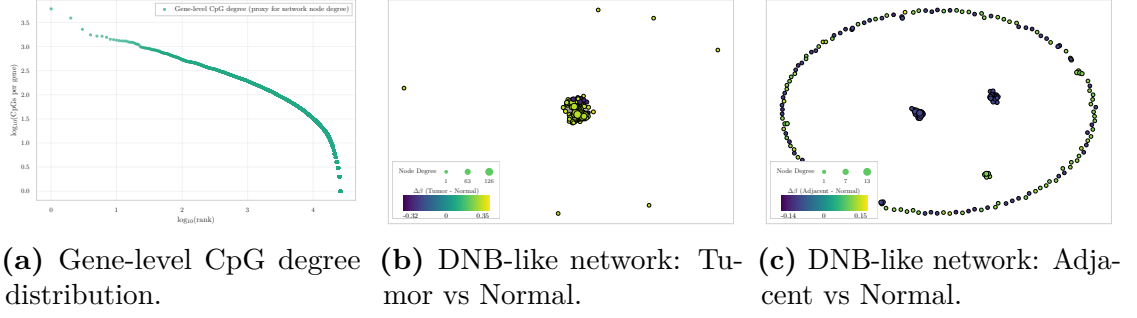


Figure 3.10: Dynamic-network proxy visualizations in GSE225845.

abundant class, followed by north and south shores in comparable proportions, whereas north and south shelves account for the smallest fractions.

This annotation profile aligns with the expected genomic composition of the EPIC array and supports correct manifest integration.

Dynamic-network proxies The degree–rank distribution (Figure 3.10a) exhibits a heavy-tailed profile, with a limited number of genes associated with many CpGs and a long tail of low-degree nodes. Such right-skewed degree structures are characteristic of heterogeneous biological networks [23].

In the Tumor vs Normal layout (Figure 3.10b), nodes aggregate into a dense and highly compact central cluster, with only a small number of isolated peripheral points. This geometry indicates that the largest Tumor–Normal $\Delta\beta$ deviations are concentrated within a cohesive subset of CpG–gene units rather than broadly distributed across the network.

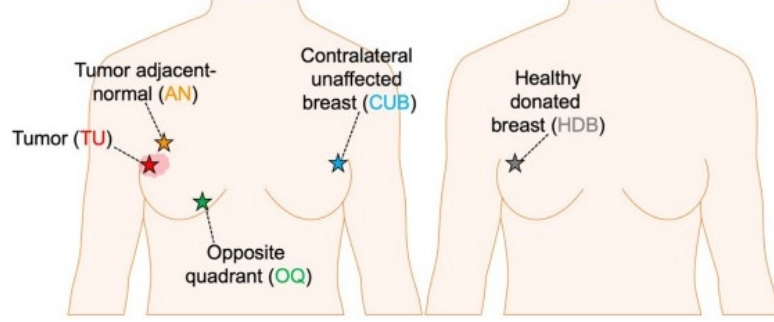
In contrast, the Adjacent vs Normal layout (Figure 3.10c) displays a markedly different configuration. While a few small central aggregates are visible, most nodes are arranged along a wide annular structure, forming a pronounced ring-like embedding. This organisation reflects weaker and more spatially dispersed deviations, without the central concentration observed in the Tumor comparison.

Altogether, the proxy visualisations indicate that Tumor–Normal contrasts generate a compact and locally coherent network signal, whereas Adjacent–Normal differences remain diffuse and geometrically fragmented.

Overall interpretation The exploratory analysis of GSE225845 indicates that, at the whole-methylome level, the three tissue states share highly similar global methylation profiles. Group-wise mean β -value densities are nearly superimposed, and both correlation structure and linear embeddings show substantial overlap. These observations indicate preservation of large-scale methylome organisation across Normal, Adjacent, and Tumor tissues, without evidence of pronounced

Table 3.3: Overview of the GSE287331 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	OQ	CUB
866,552	446	182	60	69	67	68

**Figure 3.11:** Schematic representation of the five tissue categories collected along the tumor proximity axis (TPxA), reproduced from [24].

genome-wide redistribution of methylation levels.

Locus-focused analyses reveal more structured differences. Among the 200 CpGs with the highest outlier burden, instability is predominantly directional and characterised by focal hypomethylation. Variance distributions further show increased dispersion in Tumor samples, while Adjacent tissues exhibit broader variability than Normal without forming a strictly intermediate pattern.

Nonlinear embeddings (t-SNE and UMAP) emphasise differences in spatial dispersion rather than discrete cluster separation. Normal and Adjacent samples occupy relatively compact regions, whereas Tumor samples extend over a wider area, reflecting increased heterogeneity. Dynamic-network proxies show that Tumor–Normal contrasts generate a dense central cluster of perturbed CpG–gene units, while Adjacent–Normal differences are distributed along a broader, ring-like structure without strong central concentration.

Despite the larger sample size of this cohort, which provides stable estimates of global methylation structure, the separation between Normal and Adjacent tissues remains modest. Differences between these two groups are therefore primarily focal and do not manifest as large-scale structural shifts.

Overall, GSE225845 represents a globally homogeneous dataset in which tissue-state differences become apparent only when examined through locus-specific perturbations, variability patterns, and small-scale network organisation.

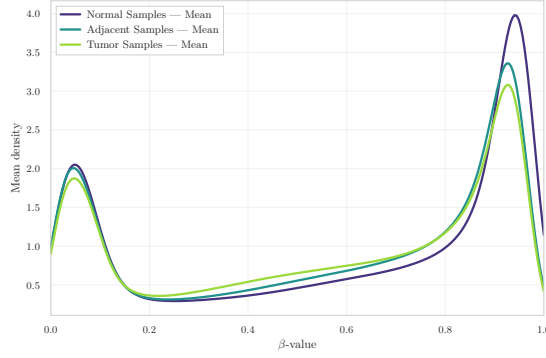


Figure 3.12: Group-wise mean β -value density in GSE287331.

3.3.3 Dataset GSE287331

Dataset overview GSE287331 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC v1.0 BeadChip. The dataset is organised along a tumor proximity axis (TPxA) spanning multiple anatomical and pathological contexts.

Five tissue categories are represented: healthy donated breast (HDB), contralateral unaffected breast (CUB), ipsilateral opposite quadrant (OQ), adjacent normal tissue (AN), and tumor (TU). An overview of the full cohort composition is reported in Table 3.3.

For cross-dataset consistency, only three biologically aligned classes are retained for downstream analyses: HDB (treated as Normal baseline), AN (Adjacent), and TU (Tumor). OQ and CUB samples are excluded, as they represent intermediate benign tissues along the proximity axis and are not directly comparable with the three-class framework adopted in the other cohorts.

Global methylation distributions The group-wise mean β -value density curves (Figure 3.12) display the expected bimodal configuration of Illumina methylation arrays. A dominant peak is observed at high methylation levels, accompanied by a smaller peak consistent with standard EPIC methylation profiles [25].

Although the overall shapes remain comparable across tissue groups, a clear ordering is visible at the high- β peak. Normal samples exhibit the highest density, Adjacent samples show a slightly attenuated peak, and Tumor samples present the lowest amplitude together with a broader distribution extending toward intermediate β values. This progressive reduction in the fully methylated peak is consistent with increasing methylation variability along the Normal–Adjacent–Tumor axis.

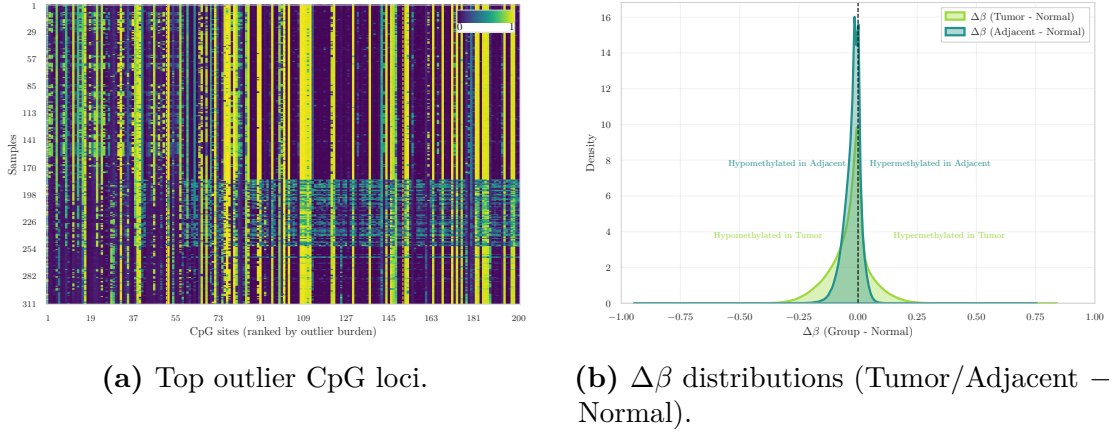


Figure 3.13: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE287331.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.13a) shows that epigenetic instability is not uniformly distributed. Alterations concentrate at specific CpG sites and within specific samples, forming vertical and horizontal bands rather than a diffuse genome-wide pattern. Both directions of deviation are observed, with focal hypermethylation and focal hypomethylation present within the selected loci [10].

The corresponding $\Delta\beta$ distributions (Figure 3.13b) highlight distinct profiles for the two contrasts. The Tumor–Normal curve is broader and displays a higher density of small negative shifts, indicating widespread but moderate hypomethylation in Tumor samples. In contrast, the Adjacent–Normal distribution is more sharply centered around zero but exhibits a comparatively more pronounced negative tail, reflecting fewer yet larger hypomethylated deviations. Overall, hypomethylation represents the dominant direction of change in both comparisons.

Variance and correlation structure The density curves (Fig. 3.14a) show that **Normal and Adjacent** samples are almost perfectly superimposed, with both groups exhibiting extremely low and tightly concentrated variance levels. Tumor samples display a *slightly* broader tail, indicating modestly increased genome-wide variability; however, the overall distribution remains sharply peaked near zero. This suggests that, at the level of CpG-wise variance, **Adjacent tissue is still epigenetically very close to Normal**, whereas Tumor exhibits only a mild elevation in variability—less pronounced than typically observed in other methylation cohorts.

A complementary perspective is provided by the sample correlation heatmap (Fig. 3.14b). As expected for genome-wide DNA methylation data, correlations are

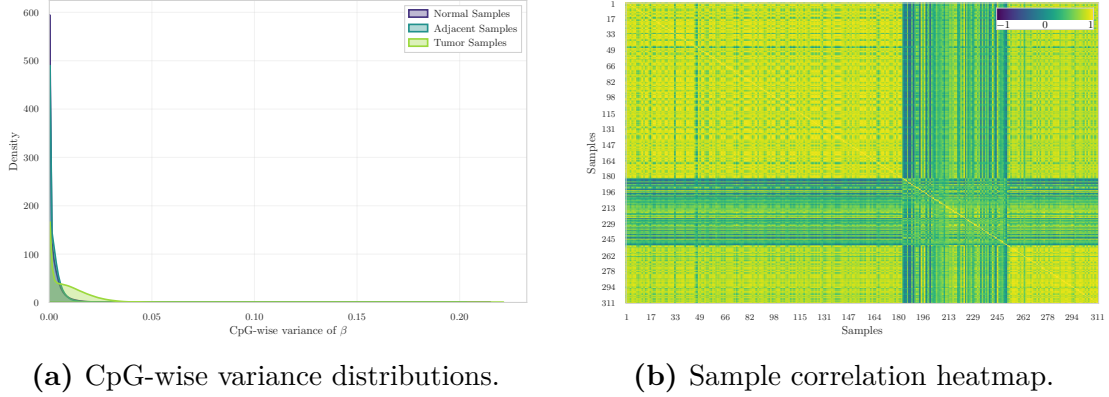


Figure 3.14: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE287331.

uniformly high across all samples, reflecting the large proportion of CpG sites that remain stable across individuals [7]. Although group labels are not displayed in the heatmap, several local blocks of high correlation are visible, suggesting structured similarity within specific subsets of samples.

Low-dimensional embeddings Dimensionality reduction using PCA, t-SNE, and UMAP was employed to visualise the global organisation of methylation profiles (Fig. 3.15). Across all three embeddings, a remarkably consistent pattern emerges: **Normal and Adjacent samples form two clearly distinct clusters**, with a sharp separation visible in PCA, strengthened in t-SNE, and appearing as a fully detached manifold in UMAP.

Tumor samples occupy a separate and more diffuse region of the embedding space, reflecting their higher epigenetic heterogeneity. Importantly, Tumor does not overlap with either Normal or Adjacent in any of the projections, further supporting the strong group structure intrinsic to this dataset.

This strong separation is also quantitatively confirmed by silhouette analysis: the average silhouette coefficient *before* BMIQ was $\text{sil}_{\text{pre}} = 0.2876$, but *after* BMIQ correction it declined to $\text{sil}_{\text{post}} = 0.2537$. Thus, BMIQ *reduced*—even if not significantly—the natural clusterability of the data, collapsing genuine biological differences—particularly the clear separation between Normal and Adjacent samples.

For this reason, and in accordance with the principle of preserving authentic biological signal, the BMIQ-normalised version was not adopted.

Genome annotation coverage The genomic distribution of annotated CpG contexts follows the characteristic architecture of the EPIC array. The majority of

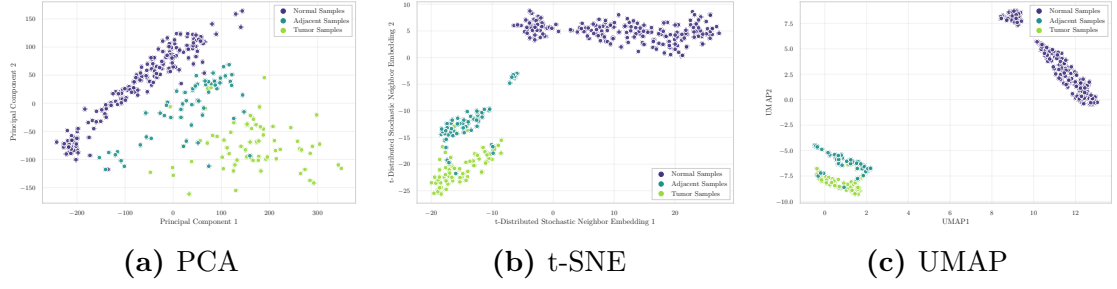


Figure 3.15: Low-dimensional embeddings of GSE287331 samples.

probes fall into **non-island (Open Sea / NA)** regions, which represent by far the largest class in the array design. This is followed by a substantial number of probes located within **CpG Islands**, and progressively smaller fractions mapping to **North Shores** and **South Shores**. As expected, **North** and **South Shelves** constitute the smallest probe categories.

This hierarchical ordering — Open Sea $\gg$ Island $>$ Shore $>$ Shelf — is fully consistent with the known structure of Infinium arrays described in the literature [18, 19].

Dynamic-network proxies The degree–rank curve (Fig. 3.16a) shows a heavy-tailed distribution of CpG counts per gene, consistent with a hub–periphery architecture typical of biological networks.

Both $\Delta\beta$ -based network layouts — Tumor vs Normal and Adjacent vs Normal (Fig. 3.16b, Fig. 3.16c) — share a similar overall structure: a **dense, compact central block** of nodes connected by short-range edges, surrounded by a ring of **peripheral, weakly connected nodes**. This common geometry reflects the underlying degree distribution and the fact that most CpG–gene units contribute minimally to group-level differences.

Despite this shared structure, the **Tumor vs Normal** comparison exhibits a more pronounced central cluster with larger $\Delta\beta$ magnitudes, suggesting a subset of CpG/gene units undergoing locally coordinated perturbations— a pattern compatible with a more unstable, transition-like network regime. In contrast, the **Adjacent vs Normal** layout shows a similarly shaped central block but with weaker deviations, indicating that adjacent tissue may experience only mild or incipient coordination of methylation changes.

Overall interpretation The GSE287331 dataset provides a clean and well-structured EPIC methylation matrix for studying breast tissue along the tumour-proximity axis. After stringent 1% probe-wise filtering and minimal median-based imputation (0.0187% NaN), the working matrix (311 samples $\times$ 702,573 CpGs) is

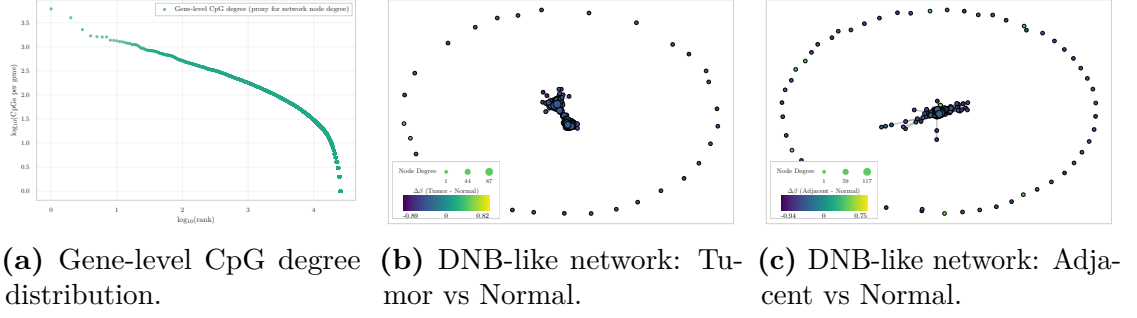


Figure 3.16: Dynamic-network proxy visualizations for GSE287331.

essentially complete. Manifest annotation confirms the expected EPIC architecture dominated by Open Sea probes, followed by CpG islands, shores, and shelves.

Exploratory diagnostics revealed that the publicly released data lacked Infinium Type I/II bias correction. BMIQ reduced probe-type discrepancies only slightly and decreased the silhouette score (from 0.2876 to 0.2537), indicating a loss of genuine biological separation. For this reason, all analyses in this thesis rely on the non-BMIQ version, which better preserves group structure.

Global methylation distributions display the typical bimodal EPIC profile with a subtle gradient (Normal > Adjacent > Tumor). Variance and correlation structures show strong similarity between Normal and Adjacent tissues and only mild variability increases in Tumor. In contrast, low-dimensional embeddings reveal clear separation of all three groups—especially between Normal and Adjacent—highlighting meaningful multivariate differences despite globally similar first- and second-order moments.

Locus-specific analyses show focused disruptions: Tumor exhibits widespread mild hypomethylation, whereas Adjacent tissue displays fewer but more pronounced deviations. Dynamic-network proxies further indicate a more compact perturbed core in Tumor samples and weaker, early-stage perturbations in Adjacent tissue, consistent with field-cancerization effects.

Overall, GSE287331 is a high-quality dataset with strong and biologically interpretable Normal–Adjacent–Tumor separation, making it well suited for downstream comparative and mechanistic analyses.

3.4 Inter Dataset Analysis

The objective of this section is not to merge the datasets—an approach that would be biologically and technically inappropriate due to differences in array platforms (HM450 vs. EPIC), preprocessing pipelines, and the strong intrinsic structure observed in GSE287331—but rather to compare their internal behaviours. This

Table 3.4: Cross-dataset comparison of Normal(N)–Adjacent(A) contrasts using global, locus-specific, and instability metrics.

Dataset	Samples A/N	CpGs	Mean $ \Delta\beta $ A–N	Var(A)/Var(N)	Outlier A/N	Silhouette A/N
GSE69914	50 / 42	485,512	0.0090	0.988	1.353	0.0028
GSE225845	113 / 140	750,426	0.0148	0.972	2.790	0.0423
GSE287331	182 / 60	702,573	0.0306	0.930	13.329	0.3442

cross-dataset perspective assesses whether the early epigenetic alterations detected in the intra-dataset analyses are *reproducible across independent cohorts*, thereby supporting the central biological premise of this thesis: detecting early methylation drift in histologically normal tissues and evaluating its potential role in tumour initiation.

3.4.1 Rationale

Across all three datasets, the intra-dataset analyses revealed three highly consistent phenomena:

- Normal and Adjacent tissues are globally similar, with nearly superimposed β -value distributions and comparable CpG-wise variance profiles.
- Tumour samples show increased heterogeneity and a clear bias toward hypomethylation, although the magnitude of this shift varies across datasets.
- Adjacent tissues exhibit early methylation drift, detectable through $\Delta\beta$ distributions and outlier profiles even when global metrics remain highly similar to Normal.

These patterns align with the expected biological progression from Normal to Tumour, in which epigenetic deregulation arises gradually and affects normal-appearing tissue surrounding the tumour. The inter-dataset analysis therefore evaluates the *directional consistency* of these effects, rather than aiming for quantitative comparability.

3.4.2 Cross-Dataset Comparative Metric

Table 3.4 reports a set of quantitative metrics designed to characterise the epigenetic deviation of Adjacent tissue from the Normal methylome across the three cohorts. Across all datasets, these metrics collectively highlight the presence of early epigenetic drift in histologically non-tumour tissue, consistent with the field-defect framework described by Teschendorff et al. [12] and recently reinforced by studies in other tissue contexts.

Mean absolute methylation difference, $\text{Mean}|\Delta\beta|(\mathbf{A}-\mathbf{N})$ For each CpG i computed the absolute mean difference between Adjacent and Normal:

$$|\Delta\beta_i| = \left| \beta_i^{(A)} - \beta_i^{(N)} \right|.$$

The cross-cohort pattern shows progressively larger locus-specific shifts from GSE69914 (0.0090) to GSE225845 (0.0148) and GSE287331 (0.0306). This monotonic increase reflects a strengthening early deviation in Adjacent tissues. Similar locus-wise methylation drifts in histologically normal tissue have been reported in independent systems, such as colonic mucosa adjacent to adenomas [26], supporting the interpretation of these alterations as early signs of field cancerization.

Global variance ratio, $\text{Var}(\mathbf{A})/\text{Var}(\mathbf{N})$ For each sample s , the global methylation variance is computed across all CpG sites:

$$v_s = \text{Var}(\beta_{s,1}, \beta_{s,2}, \dots, \beta_{s,m}),$$

where m denotes the number of CpGs. Group-wise mean variances are then obtained as

$$\bar{v}^{(N)} = \frac{1}{|N|} \sum_{s \in N} v_s, \quad \bar{v}^{(A)} = \frac{1}{|A|} \sum_{s \in A} v_s,$$

and the reported metric is their ratio:

$$\text{Var}(\mathbf{A})/\text{Var}(\mathbf{N}) = \frac{\bar{v}^{(A)}}{\bar{v}^{(N)}}.$$

Across all datasets this ratio remained consistently close to 1 (0.93–0.99), indicating that Adjacent tissue does not exhibit diffuse genome-wide instability. This aligns with recent evidence that early pre-neoplastic alterations tend to be *focal* rather than global, with only specific CpG sites showing abnormal dispersion while overall variance remains preserved [26]. Such behaviour is consistent with contemporary models of early tumourigenesis, in which subtle, locally restricted perturbations accumulate prior to large-scale chromatin deregulation and overt neoplastic transformation [27].

Outlier burden ratio, $\mathbf{A}/\mathbf{N}$ For each sample, the outlier burden is defined as the number of CpG sites whose methylation deviates from the Normal reference distribution by more than three times the Normal interquartile range:

$$\text{outlier}_s = \# \left\{ i : \left| \beta_{i,s} - \tilde{\beta}_i^{(N)} \right| > 3 \cdot IQR_i^{(N)} \right\},$$

where $\tilde{\beta}_i^{(N)}$ and $IQR_i^{(N)}$ denote the median and interquartile range of CpG i across Normal samples. The reported A/N ratio corresponds to the mean outlier burden

in Adjacent tissue divided by the mean burden in Normal tissue. This metric shows the strongest cross-dataset gradient: from mild enrichment in GSE69914 ($A/N \approx 1.35$), to moderate imbalance in GSE225845 ($A/N \approx 2.79$), to a striking amplification in GSE287331 ($A/N \approx 13.33$). These findings closely mirror evidence that increased local methylation variability in histologically normal tissue is a sensitive marker of early instability [26], and provide quantitative support for the presence of pronounced field effects in GSE287331.

Silhouette score (Normal vs. Adjacent) For each sample s , let $a(s)$ denote the mean distance from s to all other samples in the same group (cohesion), and let $b(s)$ denote the smallest mean distance from s to samples in any different group (separation). The silhouette coefficient for sample s is defined as

$$\text{sil}(s) = \frac{b(s) - a(s)}{\max\{a(s), b(s)\}},$$

and the reported score corresponds to the mean silhouette across all Normal and Adjacent samples. Higher values indicate clearer multivariate separation, whereas values near zero reflect substantial overlap.

GSE69914 (0.0028) and GSE225845 (0.0423) show almost complete overlap between Normal and Adjacent, consistent with minimal global displacement in PCA space. In contrast, GSE287331 exhibits partial separation (0.3442), indicating that the epigenetic landscape of Adjacent tissue is substantially shifted relative to Normal. This behaviour aligns with mechanistic models in which epigenetic dysregulation constitutes one of the earliest events in tumour initiation [27], preceding morphological transformation.

Together, these metrics depict a consistent Normal–Adjacent gradient across datasets. Early epigenetic drift is detectable in all cohorts, but its magnitude, focality, and multivariate visibility intensify markedly from GSE69914 to GSE287331. Importantly, the combination of stable global variance with strong outlier enrichment supports a model in which early pre-neoplastic changes arise through *localised* perturbations—a hallmark of field cancerization [26, 12].

3.4.3 Visual Comparative Analysis

To complement the quantitative summary reported in Table 3.4, this section provides a visual characterisation of the Normal–Adjacent (N–A) contrast across the three cohorts. All analyses are carried out on the three-way intersection of 326,330 CpG sites, ensuring maximal comparability across platforms (HM450 and EPIC) and preprocessing pipelines [18]. The goal is not to achieve numerical concordance at the single-CpG level—a notoriously difficult task due to platform differences, probe-design effects, and cohort-specific processing [19, 28]—but rather to evaluate

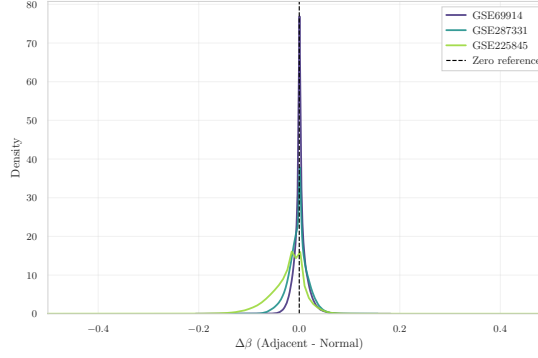


Figure 3.17: Overlay of $\Delta\beta$ (Adjacent–Normal) density curves across the three datasets, computed on the 326,330 CpG sites shared by all cohorts. All distributions are tightly centred around zero, indicating that the Adjacent–Normal shift is globally subtle yet consistent in shape across studies.

whether the *shape*, *direction*, and *global behaviour* of the early methylation drift are coherent across studies.

Global $\Delta\beta$ distributions (Adjacent–Normal) Fig. 3.17 overlays the $\Delta\beta$ (A–N) density curves for the three cohorts. All distributions are sharply centred around zero, confirming that the N–A methylation drift is globally subtle in each dataset, consistent with the focal and low-amplitude nature of early epigenetic alterations described in pre-neoplastic tissues [12, 29]. Despite differences in dispersion—reflecting platform-dependent noise, probe-type composition, and study-specific variability [16, 19]—the three curves exhibit an almost identical shape. This indicates that early epigenetic deviations in Adjacent tissue, although small, follow a *qualitatively reproducible* pattern across independent cohorts.

Pairwise $\Delta\beta$ concordance across datasets To assess whether locus-specific A–N shifts replicate across cohorts, pairwise scatterplots were generated for all dataset pairs (Fig. 3.18a, Fig. 3.18b, Fig. 3.18c). Each point corresponds to a CpG in the three-way intersection; axes show the mean $\Delta\beta$ (A–N) computed independently in each dataset.

Across all pairs, Spearman correlations were weak ($\rho = 0.05$ – 0.29), consistent with the well-established difficulty of achieving single-CpG reproducibility across independent methylation studies [28, 30, 18]. Nevertheless, the comparison between GSE69914 and GSE287331 (Fig. 3.18a) exhibits mild monotonic agreement, whereas pairs involving GSE225845 display near-zero correlation. These results indicate that while the *global* N–A drift is consistent across cohorts (as shown by the density curves), *fine-scale*, *CpG-specific* effects are heavily influenced by platform

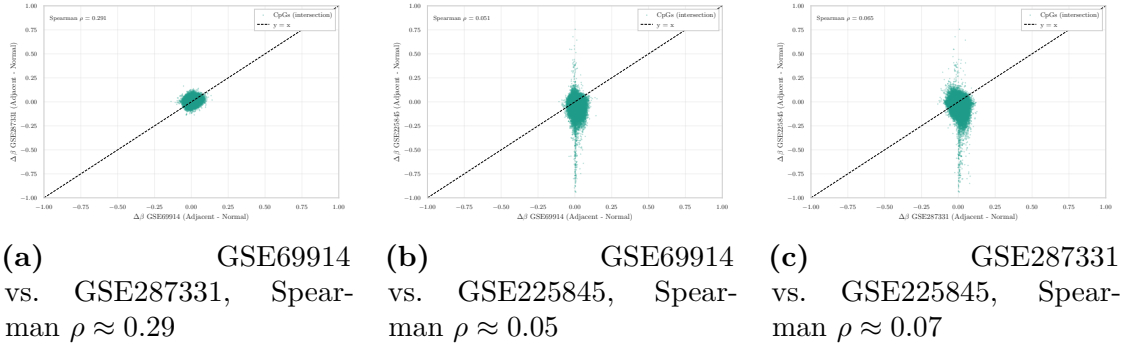


Figure 3.18: Pairwise $\Delta\beta$ (Adjacent–Normal) concordance across the three methylation datasets. (a) GSE69914 vs. GSE287331 shows a mild positive monotonic trend, (b) GSE69914 vs. GSE225845 exhibits near-zero correlation, and (c) GSE287331 vs. GSE225845 similarly shows weak agreement. Together, these comparisons highlight cohort-specific variability in locus-level methylation drift, despite stable global A–N directionality.

architecture, batch structure, and study-specific noise [19, 28].

t-SNE embeddings on the shared CpG intersection To investigate the multivariate structure of the N–A contrast across cohorts, t-SNE embeddings were computed on M-values restricted to the 326k shared CpGs. Fig. 3.19a and Fig. 3.19b display the embeddings coloured by dataset and by group, respectively.

The dataset-coloured embedding exhibits strong clustering by study, in line with the dominant influence of batch and platform effects in high-dimensional methylation data [30, 28]. This visually confirms that merging the datasets would be inappropriate, as technical structure would obscure the biological signal.

When coloured by tissue state (Normal vs. Adjacent), a subtle yet coherent within-dataset displacement becomes visible, consistent with the early, focal drift described in the epigenetic progenitor model [29, 31]. Adjacent samples tend to occupy slightly shifted regions relative to Normal samples within each dataset cluster, mirroring the quantitative metrics in Table 3.4.

Overall, the visual analyses corroborate the quantitative metrics: early epigenetic drift is detectable in all cohorts, but its magnitude and visibility vary substantially, with GSE287331 exhibiting the clearest field-defect signature. The combination of globally subtle yet directionally consistent $\Delta\beta$ patterns, weak CpG-level concordance, and dataset-specific multivariate structure reinforces the view that the Normal→Adjacent transition reflects genuine early epigenetic deregulation rather than dataset-specific artefacts.

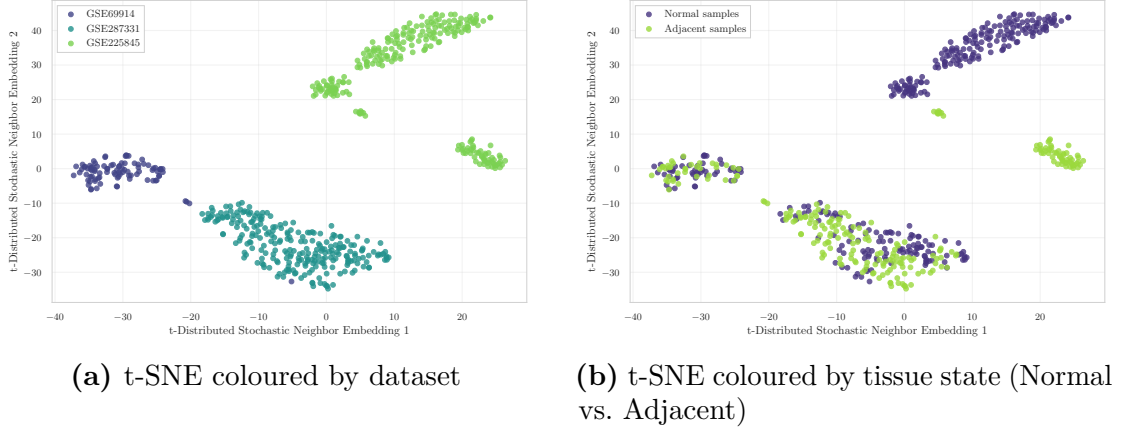


Figure 3.19: Two views of the t-SNE embedding computed on the 326k CpGs shared across GSE69914, GSE225845, and GSE287331. (a) Colouring by dataset reveals strong study- and platform-driven clustering, indicating that global methylation structure is dominated by inter-cohort effects. (b) Colouring by tissue state (Normal vs. Adjacent) shows a subtle within-dataset displacement of Adjacent samples, consistent with early localised methylation drift at the tissue margins.

3.4.4 Interpretation and implications

Taken together, the cross-dataset metrics reveal that the epigenetic deviations observed in Adjacent tissue are not idiosyncratic artefacts of a single cohort, but represent *coherent and reproducible biological signals*. Although the intensity of the effect varies across datasets, the directionality is remarkably consistent:

- Normal and Adjacent tissues remain globally similar at the whole-methylome level;
- Adjacent samples consistently exhibit early drift, detectable through locus-specific shifts, focal outlier enrichment, and mild multivariate displacement;
- Tumour samples show the expected pattern of strong hypomethylation and increased heterogeneity.

A key implication of this comparison is that the *strength* of the Normal–Adjacent signal differs substantially across cohorts. In GSE69914 and GSE225845 the drift is detectable but modest, whereas GSE287331 displays a pronounced field-defect signature across all metrics: larger locus-specific shifts, extreme outlier enrichment, and measurable separation in multivariate space. This makes GSE287331 the most informative dataset for downstream algorithmic exploration, as it provides

the strongest and clearest Normal–Adjacent contrast from which to extract early epigenetic markers.

For this reason, GSE287331 is treated as the *primary discovery dataset* for subsequent analyses, while GSE69914 and GSE225845 are used as *external validation replication cohorts*. Importantly, in this work the term “validation” refers to *biological* replication across independent datasets—rather than machine-learning validation. The discovery step is carried out in GSE287331, and the resulting CpG markers or patterns are then evaluated in the two additional cohorts to test whether the same loci display consistent Normal–Adjacent drift across different platforms, sample compositions, and preprocessing pipelines.

Because the three cohorts differ in array technology, processing pipelines, and study design, merging them would not be biologically meaningful; instead, the inter-dataset analysis serves as a multi-cohort consistency check, supporting the interpretation that the Normal→Adjacent drift represents genuine early epigenetic deregulation rather than dataset-specific noise. This framework aligns with contemporary models of tumour initiation, in which subtle, locally restricted perturbations arise in histologically normal tissue prior to overt neoplastic transformation.

3.5 Motivation for next Data Pre-Processing

Chapter 4

Data preprocessing

REPORT 02

Missingness filtering and imputation I quantified the amount and structure of missing values in the β -matrix. Probe-wise missingness is a well-known source of unreliability in Illumina methylation arrays, and several studies recommend removing CpGs that exhibit low measurement quality or an excessive proportion of missing values. For example, Islam *et al.* [32] excluded probes with missing β -values in more than 2% of samples in their pediatric tissue methylation analysis, while Mansell *et al.* [33] applied a similar 1% threshold for detection failure in their EPIC-array quality control pipeline.

Guided by this evidence and the large number of CpG sites available, I adopted a stringent probe-wise filtering criterion and removed all CpGs missing in more than 1% of samples. This choice prioritises high-confidence measurements while minimising reliance on imputed values.

After filtering, only a negligible fraction of values remained missing in the working matrix (58 561 NaN, corresponding to 0.0187% of all entries). Given this extremely low level of sparsity, median-based imputation is both statistically robust and distribution-preserving. Following current recommendations, where missing methylation values are replaced by probe-wise medians to avoid introducing group-specific bias, all residual NaN were imputed using the median β -value of each CpG across samples. This approach is consistent with recent harmonization methods such as mLiftOver, where missing methylation values are explicitly replaced using probe-wise median substitution to preserve biologically plausible levels [34].

Evidence of Missing Infinium Type I/II Probe Bias Correction During the exploratory analysis performed uniformly across all three datasets, the per-sample mean β -value distributions (Fig. 4.1a, Fig. 4.1b, Fig. 4.1c) revealed a marked

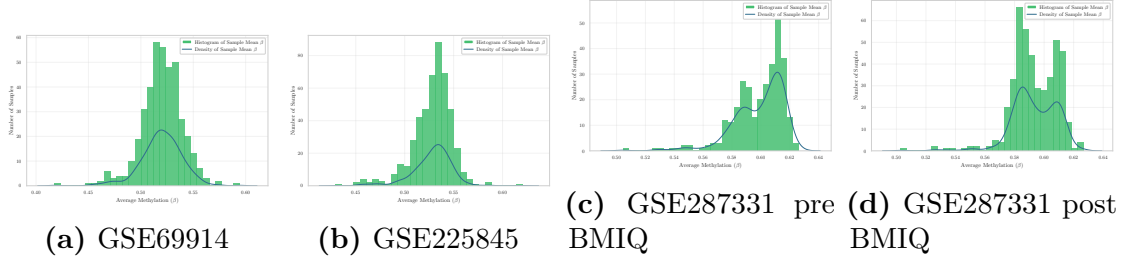


Figure 4.1: Per-sample mean β distributions for the three datasets. **GSE69914** and **GSE225845** show a single broad peak, whereas **GSE287331** displays a right-shifted bimodal pattern. After BMIQ, the two peaks become slightly closer and more similar in height, but the bimodality persists.

inconsistency. Whereas **GSE69914** and **GSE225845** displayed a single broad peak of global methylation levels, **GSE287331** showed a much narrower distribution, shifted towards higher β -values and characterized by two distinct modes in the high- β range.

This unusual right-shifted bimodality—largely absent in the other datasets—strongly indicated that a key preprocessing step, namely Infinium Type I/II probe-design bias correction, had not been applied in the processed data released by the authors.

To confirm this hypothesis, I stratified all probes by Infinium design type (Type I vs. Type II) using the official Illumina EPIC v1.0 manifest, obtained from the corresponding Bioconductor annotation package [35]. Using this manifest, the β -value distributions were separated by probe design and inspected through a dedicated diagnostic plot (Fig. 4.2). According to the expected behaviour described by Teschendorff et al. [7], datasets that have undergone BMIQ normalization should display nearly overlapping β -value distributions for the two probe types.

In contrast, the Q–Q plot comparing Type I and Type II probes revealed a markedly elevated median absolute deviation (median $|\Delta| = 0.380$), indicating a strong and systematic quantile-wise shift between the two designs. Such a deviation is fully incompatible with BMIQ-corrected data, which should instead exhibit deviations close to zero across all quantiles. After applying BMIQ, this deviation decreases only marginally ($|\Delta| : 0.380 \rightarrow 0.370$), confirming that the correction improves probe-type alignment but only to a limited extent.

This modest improvement is also clearly visible in the diagnostic density plots (Fig. 4.2b). After BMIQ, the rightmost portion of the distribution (near $\beta \approx 1$) shows Type II probes becoming visibly more similar to Type I, and the strong asymmetry between the two curves is partially reduced. Nevertheless, the overall shapes of the two probe distributions remain distinct, and the characteristic “double-tail” pattern of the uncorrected dataset is still present after normalization. BMIQ

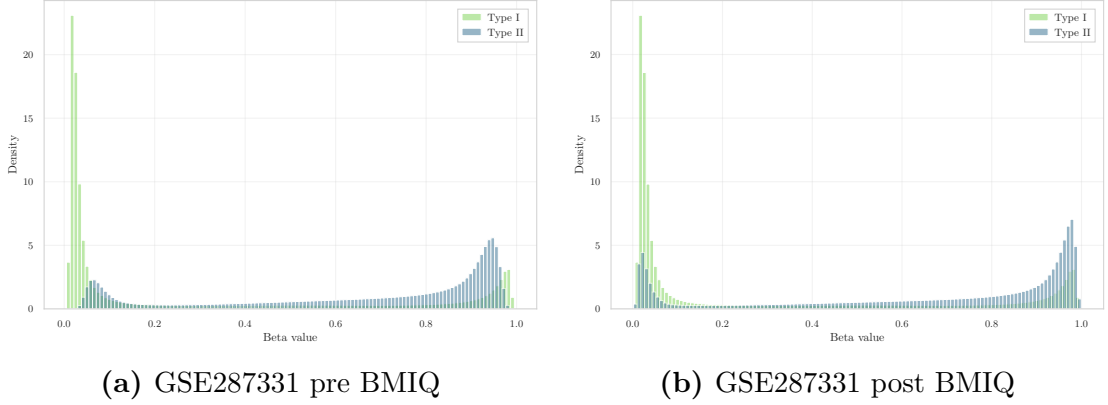


Figure 4.2: Type I vs. Type II probe design comparison before and after BMIQ. After correction, Type II probes become more similar to Type I in the high- β region (especially near $\beta \approx 1$), and the overall discrepancy is reduced ($|\Delta| : 0.380 \rightarrow 0.370$), though probe types remain clearly distinct.

therefore warps the distribution in the expected direction, but the magnitude of the adjustment is small and insufficient to reconcile the two probe types to the degree observed in datasets processed with a full bias-correction pipeline.

A similar behaviour can be observed in the per-sample mean β distributions: after BMIQ, the two peaks become slightly higher and closer to one another (Fig. 4.1d), indicating that the algorithm partially reduces the gap between the two dominant modes. However, even after correction, the distribution remains visibly and consistently *bimodal*, suggesting that BMIQ mitigates only weakly the underlying discrepancy and that sample-level heterogeneity continues to dominate the global methylation structure.

Taken together, these observations provide clear visual evidence that no probe-design bias correction was applied in the original processing. To further support this conclusion, I examined the official publication [24]. The paper carefully describes the acquisition, processing, and analytical use of Illumina EPIC β -values, but it does *not* mention BMIQ, SWAN, noob, or any other method aimed at correcting Infinium probe-type bias. The absence of such a statement—combined with the strong visual mismatch between Type I and Type II probe distributions—confirms that no probe-design bias correction was applied before the release of the dataset.

For these reasons, applying a dedicated Infinium Type I/II bias correction step is necessary before performing any comparative analysis.

BMIQ correction: implementation and methodological considerations

Given the clear evidence of missing Infinium Type I/II probe-bias correction, I applied the canonical BMIQ algorithm [7], using the implementation provided

by the `watermelon` package [36], to harmonize the distributions of Type I and Type II probes for each sample. To ensure full reproducibility and strict memory control on the extremely wide EPIC matrix, I implemented a *sequential, sample-wise BMIQ correction*, avoiding any between-sample pooling or multi-core parallelization. Each sample’s β vector was processed independently using a safe wrapper around `watermelon::BMIQ`, which first verified the availability of a sufficient number of non-missing Type I and Type II probes and then returned the corrected β -values. The corrected matrix was updated in place, preserving the schema `id_tissue | CpGs... | label` and subsequently written to a single Parquet file. This design guarantees that no cross-sample information influenced the correction and that the warping applied by BMIQ reflects exclusively the distributional characteristics of each individual sample.

While BMIQ is widely used to mitigate probe-design bias, multiple studies have raised important methodological caveats regarding normalization procedures that operate on the full distribution of methylation values. Yousefi *et al.* [37] highlighted that quantile-based adjustments and related normalization strategies may “come at the cost of *artificially reduced biological variability*,” a phenomenon that is particularly undesirable when studying biological groups expected to differ globally in their methylation profiles. Similarly, Liu *et al.* [38] reported that, when methylation alterations are extensive in size and number, certain normalization methods can “*remove biological signal*,” effectively shrinking meaningful differences between groups.

More recently, Welsh *et al.* [39] performed a systematic evaluation of normalization methods for EPIC arrays and showed that approaches enforcing strong distributional alignment may distort real biological structure. In particular, the authors observed that quantile-based normalization “*forces the empirical marginal distributions of the samples to be the same, removing all variation in this statistic*,” and reported that BMIQ, in some settings, yielded “*distorted and discontinuous*” distributions. Such findings provide a conceptual explanation for the behaviour observed: although BMIQ was applied strictly per-sample, its within-array warping can attenuate global differences between Normal, Adjacent, and Tumor samples when these differences are manifested as broad shifts in the underlying β -value distribution.

Together, these considerations emphasise both the necessity and the limitations of probe-design bias correction. Applying BMIQ was essential to remove the technical Type I/II bias absent in the original processing, yet its known tendency to reduce distributional heterogeneity across groups must be carefully taken into account when interpreting group-level contrasts in downstream analyses.

Importantly, the modest degree of probe-type harmonization is accompanied by a noticeable flattening of biological structure: as will be shown in the low-dimensional embedding analysis (Section 3.3.3), PCA, UMAP, and t-SNE projections reveal

that the Normal, Adjacent, and Tumor groups become *less clearly separable* after BMIQ. For this reason, all downstream analyses are carried out on the **non-BMIQ version** of the dataset, which preserves sharper cluster boundaries and better reflects the underlying biological signal.

Chapter 5

Feature selection

REPORT 03

Chapter 6

Model training and evaluation

REPORT 04

Chapter 7

Dynamic System for biomarker detection

REPORT 05

Chapter 8

Conclusion and future work

CONCLUSIONI E PROSPETTIVE DI FUTURO

Chapter 9

Appendix

AGGIUNGERE APPENDICE

Bibliography

- [1] P. Yousefi, K. Huen, R. Aguilar Schall, A. Decker, E. Elboudwarej, H. Quach, L. Barcellos, and N. Holland. «Considerations for Normalization of DNA Methylation Data by Illumina 450K BeadChip Assay in Population Studies». In: *Epigenetics* 8.11 (Aug. 2013), pp. 1141–1152. DOI: 10.4161/epi.26037 (cit. on p. 3).
- [2] National Center for Biotechnology Information (NCBI). *Gene Expression Omnibus (GEO)*. Public functional genomics data repository supporting MIAME-compliant submissions and providing tools to query and download curated experiments. NCBI. n.d. URL: <https://www.ncbi.nlm.nih.gov/geo/> (cit. on p. 4).
- [3] National Center for Biotechnology Information (NCBI). *GSE69914: Genome wide DNA methylation profiling of normal breast, normal adjacent and breast cancer tissue*. Gene Expression Omnibus (GEO). 2015. URL: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE69914> (cit. on p. 4).
- [4] National Center for Biotechnology Information (NCBI). *GSE225845: High neighborhood deprivation impacts DNA methylation and gene expression in cancer-related genes*. Gene Expression Omnibus (GEO). 2023. URL: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE225845> (cit. on p. 4).
- [5] National Center for Biotechnology Information (NCBI). *GSE287331: DNA methylation patterns in breast cancer, paired benign tissue from ipsilateral and contralateral breast, and healthy controls*. Gene Expression Omnibus (GEO). 2025. URL: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE287331> (cit. on p. 5).
- [6] P. Du, X. Zhang, C. Huang, N. Jafari, W. A Kibbe, L. Hou, and S. M Lin. «Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis». In: *BMC Bioinformatics* 11.1 (2010), p. 587. DOI: 10.1186/1471-2105-11-587 (cit. on p. 5).

- [7] Andrew E. Teschendorff, Francesco Marabita, Matthias Lechner, Thomas Bartlett, David Tegner, Jonathan Gomez-Cabrero, and Stephan Beck. «A beta-mixture quantile normalization method for correcting probe design bias in Illumina Infinium 450k DNA methylation data». In: *Bioinformatics* 29.2 (2013), pp. 189–196. DOI: 10.1093/bioinformatics/bts680. URL: <https://academic.oup.com/bioinformatics/article/29/2/189/199637> (cit. on pp. 5, 9, 20, 31, 32).
- [8] B. P. de Almeida, G. Richard, H. Dalla-Torre, C. Blum, L. Hexemer, P. Pandey, S. Laurent, C. Rajesh, M. Lopez, A. Laterre, M. Lang, U. Sahin, K. Beguir, and T. Pierrot. «A Multimodal Conversational Agent for DNA, RNA and Protein Tasks». In: *Nature Machine Intelligence* (2025). DOI: 10.1038/s42256-025-01047-1 (cit. on p. 5).
- [9] A. V. Sokolov and H. B. Schiöth. «Decoding Depression: A Comprehensive Multi-cohort Exploration of Blood DNA Methylation Using Machine Learning and Deep Learning Approaches». In: *Translational Psychiatry* 14.1 (2024). DOI: 10.1038/s41398-024-02992-y (cit. on p. 6).
- [10] M. Ehrlich. «DNA methylation in cancer: Too much, but also too little». In: *Oncogene* 21 (Aug. 2002), pp. 5400–5413. DOI: 10.1038/sj.onc.1205651 (cit. on pp. 7, 8, 19).
- [11] A. Bird. «DNA methylation patterns and epigenetic memory». In: *Genes & Development* 16.1 (Jan. 2002), pp. 6–21. DOI: 10.1101/gad.947102 (cit. on pp. 7, 8).
- [12] A. E. Teschendorff, Y. Gao, A. Jones, M. Ruebner, M. W. Beckmann, D. L. Wachter, P. A. Fasching, and M. Widschwendter. «DNA methylation outliers in normal breast tissue identify field defects that are enriched in cancer». In: *Nature Communications* 7 (Jan. 2016), p. 10478. DOI: 10.1038/ncomms10478 (cit. on pp. 7–9, 23, 25, 26).
- [13] National Cancer Institute. *BRCA Gene Changes: Cancer Risk and Genetic Testing*. <https://www.cancer.gov/about-cancer/causes-prevention/genetics/brca-fact-sheet>. Accessed October 2025. Oct. 2024 (cit. on p. 7).
- [14] T. E. Bartlett, I. Evans, A. Jones, J. E. Barrett, S. Haran, D. Reisel, K. Papaikonomou, L. Jones, C. Herzog, N. Pashayan, B. M. Simoes, R. B. Clarke, D. G. Evans, T. S. Ghezelayagh, S. Ponandai-Srinivasan, N. R. Boggavarapu, P. G. Lalitkumar, S. J. Howell, R. A. Risques, A. F. Rådestad, L. Dubeau, K. Gemzell-Danielsson, and M. Widschwendter. «Antiprogestins reduce epigenetic field cancerization in breast tissue of young healthy women». In: *Genome Medicine* 14.1 (June 2022). DOI: 10.1186/s13073-022-01063-5 (cit. on p. 7).

- [15] K. P. M. Suijkerbuijk, M. J. Fackler, S. Sukumar, C. H. van Gils, T. van Laar, E. van der Wall, M. Vooijs, and P. J. van Diest. «Methylation is less abundant in BRCA1-associated compared with sporadic breast cancer». In: *Annals of Oncology* 19 (Nov. 2008), pp. 1870–1874. DOI: 10.1093/annonc/mdn409 (cit. on p. 7).
- [16] J. Maksimovic, L. Gordon, and A. Oshlack. «SWAN: Subset-Quantile Within Array Normalization for Illumina Infinium HumanMethylation450 Bead-Chips». In: *Genome Biology* 13.6 (2012), R44. DOI: 10.1186/gb-2012-13-6-r44 (cit. on pp. 8, 26).
- [17] *Infinium MethylationEPIC v1.0 B5 Manifest File*. https://support.illumina.com/array/array_kits/infinium-methylationepic-beadchip-kit/downloads.html. Illumina, accessed 2025 (cit. on p. 10).
- [18] W. Zhou, P. W. Laird, and H. Shen. «Comprehensive characterization, annotation and innovative use of Infinium DNA methylation BeadChip probes». In: *Nucleic Acids Research* 45.4 (2017), e22. DOI: 10.1093/nar/gkw967 (cit. on pp. 11, 21, 25, 26).
- [19] R. Pidsley, E. Zotenko, T. J. Peters, M. G. Lawrence, G. P. Risbridger, P. Molloy, S. Van Dijk, B. Muhlhausler, C. Stirzaker, and S. J. Clark. «Critical evaluation of the Illumina MethylationEPIC BeadChip microarray for whole-genome DNA methylation profiling». In: *Genome Biology* 17 (2016), p. 208. DOI: 10.1186/s13059-016-1066-1 (cit. on pp. 11, 21, 25–27).
- [20] L. Chen, R. Liu, Z.-P. Liu, M. Li, and K. Aihara. «Detecting early-warning signals for sudden deterioration of complex diseases by dynamical network biomarkers». In: *Scientific Reports* 2 (2012), p. 342. DOI: 10.1038/srep00342 (cit. on p. 11).
- [21] J. West, G. Bianconi, S. Severini, and A. E. Teschendorff. «On dynamic network entropy in cancer». In: *arXiv:1202.3015* (2018). URL: <https://arxiv.org/abs/1202.3015> (cit. on p. 11).
- [22] J. Liu, D. Ding, J. Zhong, and R. Liu. «Identifying the critical states and dynamic network biomarkers of cancers based on network entropy». In: *Journal of Translational Medicine* 20.254 (2022). DOI: 10.1186/s12967-022-03445-0 (cit. on p. 11).
- [23] P. Wang. «Network biology: Recent advances and challenges». In: *Gene & Protein in Disease* 1.2 (2022), p. 101. DOI: 10.36922/gpd.v1i2.101 (cit. on p. 16).

- [24] S. R. Dennis, T. Tsukioki, G. Cottone, W. Zhou, P. A. Ganz, M. E. Sehl, Y. Luo, S. A. Khan, and S. Clare. «DNA methylation patterns in breast cancer, paired benign tissue from ipsilateral and contralateral breast, and healthy controls». In: *Breast Cancer Research* 27.1 (2025), p. 103. DOI: 10.1186/s13058-025-02057-y (cit. on pp. 17, 32).
- [25] J. Maksimovic, B. Phipson, and A. Oshlack. «A cross-package Bioconductor workflow for analysing methylation array data». In: *F1000Research* 5 (2017), p. 1281. DOI: 10.12688/f1000research.8839.3 (cit. on p. 18).
- [26] J. Yates, H. Schaufelberger, R. Steinacher, P. Schär, K. Truninger, and V. Boeva. «DNA-methylation variability in normal mucosa: A field cancerization marker in patients with adenomatous polyps». In: *Journal of the National Cancer Institute* 116.6 (2024), pp. 974–982. DOI: 10.1093/jnci/djae016 (cit. on pp. 24, 25).
- [27] S. Zhang, X. Xiao, Y. Yi, X. Wang, L. Zhu, Y. Shen, D. Lin, and C. Wu. «Tumor initiation and early tumorigenesis: Molecular mechanisms and epigenetic drivers». In: *Signal Transduction and Targeted Therapy* (2024). DOI: 10.1038/s41392-024-01848-7 (cit. on pp. 24, 25).
- [28] J.-P. Fortin, A. Labbé, M. Lemire, B. W. Zanke, T. J. Hudson, E. J. Fertig, C. M. T. Greenwood, and K. D. Hansen. «Functional normalization of 450k methylation array data improves replication in large cancer studies». In: *Genome Biology* 15.12 (2014), p. 503. DOI: 10.1186/s13059-014-0503-2 (cit. on pp. 25–27).
- [29] W. Timp and A. P. Feinberg. «Cancer as a dysregulated epigenome: The epigenetic progenitor model of cancer». In: *Nature Reviews Cancer* 13.7 (2013), pp. 497–510. DOI: 10.1038/nrc3486 (cit. on pp. 26, 27).
- [30] J. T. Leek, R. B. Scharpf, H. Corrada Bravo, D. Simcha, B. Langmead, W. E. Johnson, D. Geman, K. Baggerly, and R. A. Irizarry. «Tackling the widespread and critical impact of batch effects in high-throughput data». In: *Nature Reviews Genetics* 11.10 (2010), pp. 733–739. DOI: 10.1038/nrg2825 (cit. on pp. 26, 27).
- [31] K. D. Hansen, W. Timp, H. Corrada Bravo, S. Sabunciyan, B. Langmead, O. G. McDonald, B. Wen, H. Wu, Y. Liu, D. Diep, E. Briem, K. Zhang, R. A. Irizarry, and A. P. Feinberg. «Increased methylation variation in epigenetic domains across cancer types». In: *Nature Genetics* 43 (2011), pp. 768–775. DOI: 10.1038/ng.865 (cit. on p. 27).

- [32] S. A. Islam, S. J. Goodman, J. L. MacIsaac, J. Obradović, R. G. Barr, W. T. Boyce, and M. S. Kobor. «Integration of DNA methylation patterns and genetic variation in human pediatric tissues helps inform EWAS design and interpretation». In: *Epigenetics & Chromatin* 12.1 (2019), p. 1. DOI: 10.1186/s13072-018-0245-6 (cit. on p. 30).
- [33] G. Mansell, T. J. Gorrie-Stone, Y. Bao, M. Kumari, L. S. Schalkwyk, J. Mill, and E. Hannon. «Guidance for DNA methylation studies: Statistical insights from the Illumina EPIC array». In: *BMC Genomics* 20.366 (2019), pp. 1–15. DOI: 10.1186/s12864-019-5761-7 (cit. on p. 30).
- [34] B. H. Chen and W. Zhou. «mLiftOver: Mapping DNA methylation data across array platforms and genome builds». In: *Bioinformatics* 40.7 (2024), btae423. DOI: 10.1093/bioinformatics/btae423 (cit. on p. 30).
- [35] M. D. Robinson, G. K. Smyth, and K. D. Hansen. *IlluminaHumanMethylation450kanno.ilmn12.hg19: Annotation for Illumina’s 450k methylation arrays*. R package version 0.6.0. Bioconductor. 2015. URL: <https://bioconductor.org/packages/IlluminaHumanMethylation450kanno.ilmn12.hg19> (cit. on p. 31).
- [36] R. Pidsley, C. C. Y. Wong, M. Volta, K. Lunnon, J. Mill, and L. C. Schalkwyk. «A data-driven approach to preprocessing Illumina 450K methylation array data». In: *BMC Genomics* 14.293 (2013), pp. 1–10. DOI: 10.1186/1471-2164-14-293 (cit. on p. 33).
- [37] P. Yousefi, K. Huen, R. Aguilar Schall, A. Decker, E. Elboudwarej, H. Quach, L. Barcellos, and N. Holland. «Considerations for normalization of DNA methylation data by Illumina 450K BeadChip assay in population studies». In: *Epigenetics* 8.11 (2013), pp. 1145–1155. DOI: 10.4161/epi.26037 (cit. on p. 33).
- [38] J. Liu and K. D. Siegmund. «An evaluation of processing methods for Illumina HumanMethylation450 BeadChip platform». In: *BMC Genomics* 17.469 (2016). DOI: 10.1186/s12864-016-2819-7 (cit. on p. 33).
- [39] H. Welsh, C. M. P. F. Batalha, W. Li, K. L. Mpye, N. C. Souza-Pinto, M. S. Naslavsky, and E. J. Parra. «A systematic evaluation of normalization methods and probe replicability using Infinium EPIC 850K methylation data». In: *Clinical Epigenetics* 15.1 (2023), p. 59. DOI: 10.1186/s13148-023-01459-z (cit. on p. 33).